

SUPPLY & DEMAND PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

SPRING 2026

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Problem 1

The chocolate market in Kazakhstan is represented by the following equations of demand and supply:

$$Q_{\text{choc},D} = 560 - 4P_{\text{choc}} + 0.1Y + 4P_{\text{caramel}} - 5P_{\text{milk}} \quad (1)$$

$$Q_{\text{choc},S} = -60 + 6P_{\text{choc}} - 10P_{\text{cacao}} \quad (2)$$

Where:

- $Q_{\text{choc},D}$ is the quantity of chocolate demanded, in kilograms.
- $Q_{\text{choc},S}$ is the quantity of chocolate supplied, in kilograms.
- P_{choc} is the price per kilogram of chocolate.
- P_{caramel} is the price of caramels, and is 50 tenge.
- P_{milk} is the price of milk, and is 8 tenge.
- P_{cacao} is the price of the cacao grain, and is 10 tenge.
- w is the wage rate per hour worked.
- L is the number of hours worked.

$Y = w \times L$ is the consumers' income, which is obtained by working in the labor market, represented by the following equations of demand and supply:

$$L_D = 100 - 2w \quad (3)$$

$$L_S = -20 + 4w \quad (4)$$

Compute the equilibrium.

Problem 2

Consider two villages that sell honey. In the first village, the demand for honey is:

$$Q_1^d = 200 - P \quad (5)$$

and the supply of honey is:

$$Q_1^s = -40 + 5P \quad (6)$$

In the second village, demand for honey is:

$$Q_2^d = 100 - 2P \quad (7)$$

and supply of honey is:

$$Q_2^s = 20 + 4P \quad (8)$$

- (a) Calculate equilibrium on the honey market in each village. Draw the graphs of supply, demand and equilibria for each village.
- (b) Suppose a rising entrepreneur *Panqueque* decides to earn some money by buying some honey for a lower price in one of the villages and reselling it for a higher price in another village. He decides to buy 30 bottles of honey from the village with the lower price. Show (algebraically) how this decision influences demand and/or supply of this village. Calculate the new equilibrium price.
- (c) Panqueque resells the 30 bottles of honey in the second village. Show (algebraically) how this decision influences demand and/or supply in this village. Calculate the new equilibrium price.
- (d) How much money did Panqueque get from his enterprise?
- (e) Now find the optimal quantity that Panqueque should have bought from the village with lower price and then resold in the village with higher price if he wanted to get the most profit from this enterprise.

Problem 3: Graphical exercises

- (a) The market of Kumis was initially in equilibrium. A health report finds that Shubat (a substitute to Kumis) is bad for the health. At the same time, the wages of horse care workers increase due to changes in the labor market. Show on the graph what happens to demand/supply of Kumis and to Kumis's equilibrium price/quantity.
- (b) The market of clay plates was initially in equilibrium. Due to technological advances, the clay became cheaper. Show on the graph what happens to supply/demand on the market and to equilibrium price/quantity.
- (c) The market for cat food is initially in equilibrium and there are two firms producing the cat food. Suppose that two new firms enter the cat food market. Show on the graph what happens to demand/supply on the cat food market and equilibrium price/quantity.
- (d) The market of bus rides is initially in equilibrium. A health report shows that walking instead of taking a bus has a positive effect on the environment and also improves health. Show on the graph what happens to supply/demand on the market and to equilibrium price/quantity on the market.